

PlantCare: Scan, Identify, and Grow

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Project Overview

PlantCare is a multi-screen native iOS application designed to help users identify plant species from photographs and learn how to care for them. Users can photograph plants via the iPhone camera or select photos from their gallery. The application communicates with live botanical REST APIs to display candidate species, confidence scores, and visible plant condition suggestions. Users can save identified plants, photos, and care guides to a personal "My Garden" collection retained across launches. Target users include beginner gardeners, houseplant owners, and students.

Main Application Screens

1. Sign In & Sign Up: Onboarding & authentication gateway supporting Sign in with Apple (Face ID/Touch ID), email/password login, plant parent profile personalization, and an Apple HIG-compliant "Continue as Guest" mode for zero-friction exploration.
2. Home & Explore: Features an "Explore Plants" search bar, curated botanical species carousel, and quick plant care tips.
3. Scan Plant: Clean botanical camera viewfinder with corner framing accents, native photo picker, and photo preview.
4. Identification Result: Displays photo thumbnail, primary matched species, confidence progress meter (0–100%), and alternative candidates.
5. Plant Profile and Care: Detailed care parameters (watering schedule, sunlight, growth cycle) and non-diagnostic condition advice, with an animated "Save to My Garden" action.
6. My Garden: A persistent digital collection displaying saved plant photos, common names, dates added, and condition badges with swipe-to-delete.

External API Integration

The application utilizes the Pl@ntNet identification API to analyze photographs via multipart form uploads, returning likely species, scientific taxonomy, and confidence scores. After identification, the scientific name is used to query the Perennial Plant API for rich care instructions (watering, sunlight, and growth cycles). Implemented with URLSession, async/await concurrency, and Codable models with comprehensive error handling for offline states, rate limits, and network dropouts. API keys are shielded via .xcconfig templates outside public version control.

Local Data Persistence & Security

SwiftData (@Model SavedPlant) stores plant metadata, care metrics, condition summaries, and dates. Captured photos are saved directly into the application's sandboxed Documents directory. User authentication credentials and tokens are secured inside the hardware-encrypted iOS Keychain (kSecClassGenericPassword). During the live demonstration, the team will save a plant, force-close the app process completely, and relaunch it to prove that all data and images remain stored.

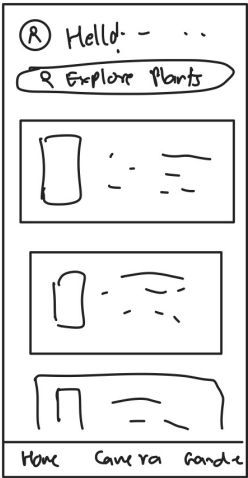
Advanced User Interface

PlantCare incorporates a Liquid Glass / Glassmorphism design system using SwiftUI materials (.ultraThinMaterial), glowing gradient accents, an Apple Camera-inspired botanical viewfinder, fluid spring animations, tactile haptic feedback, and Apple HIG compliance.

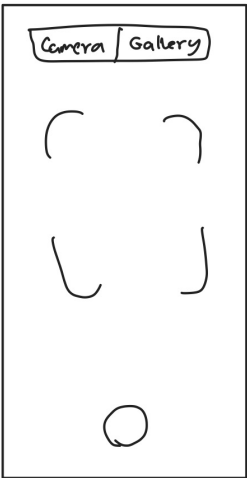
App Screen Hand-Drawn Wireframe Sketches

Screens: Sign In/Guest, Home View, Scan Viewfinder, Plant Details & Profile, and My Garden

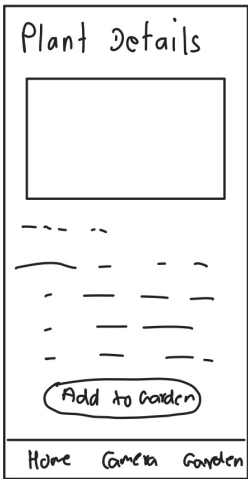
Home



Analysis



Plant



Garden

